

## REPORT ON ONLINE RETAIL STORE (UK-Based Store)

The analysis explored the purchases made for an online retail company based in the UK during an eight-month period. The goal was to understand revenue, what goods brought about the highest revenue, what goods were mostly returned and how sales was every month.

For **overall revenue performance**, the business generated a huge revenue with evidence of seasonal variation. Revenue peaked during the ember months, starting September and recording the highest in November with a slight drop in December. This pattern lets us know that seasonal demand cycles such as holiday shopping and promotional periods helped with revenue.

Upon further analysis, I realized that for **geographic performance**, the business made most sales in the United Kingdom, which accounted for over £8M. Other countries such as Netherlands, EIRE, Germany, France, Australia, Switzerland contributed less than £300K each. This helps us understand that there is limited international market penetration and an opportunity for global expansion.

For **product performance**, a small number of products generated a large share of revenue and revenue wasn't evenly distributed across the product catalog. This suggests that the business heavily relies on certain products, necessitating a need for strategy.

Further analysis made me realize **returns** were huge part of the dataset. Return losses are concentrated in a small set of products and certain items consistently contribute to negative revenue. This highlights that there might be a problem with product quality or a case of product mismatch or false advertising.

**Customer analysis** showed that revenue is concentrated among top customers, with the top 10 customers contributing approximately 13% to the total revenue. This suggests that there's no extreme dependence on a single buyer and there's a need to retain high-value clients for stability.

**For relationship between variables**, there's a strong positive correlation between revenue and quantity meaning higher purchase volumes drive high revenue and a negative relationship between unit price and quantity, suggesting that higher-priced items are usually purchased in smaller quantities.

Several limitations were identified: Missing CustomerID reduced the impact of customer behavior analysis, certain non-product entries were present (Amazon fees, discount, commissions, postage) were present and required filtering and some transactions could not be attributed to identifiable customers.